FORCE AND PRESSURE -03

Class 08 - Science

1. What is pressure? **[1]**
2. What kind of force is an electrostatic force? **[1]**
3. A rocket has been fired upwards to launch a satellite in its orbit. Name the two forces acting on the rocket immediately after leaving the launching pad. **[1]**
4. What is gravitational force? **[1]**
5. Name some non-contact forces. **[1]**
6. On the basis of interaction what are the two categories of forces? **[1]**
7. When we press the bulb of a dropper with its nozzle kept in water, air in the dropper is seen to escape in the form of bubbles. Once we release the pressure on the bulb, water gets filled in the dropper. The rise of water in the dropper is due to - **[1]**
   (a) pressure of water
   (b) gravity of the earth
   (c) shape of rubber bulb
   (d) atmospheric pressure.
8. Does liquids also exert pressure? **[1]**
9. What is atmosphere? **[1]**
10. How can we increase the pressure by exerting same force? **[1]**
11. Can you separate two hemispheres, if all the air is suck out from them? **[1]**
12. What do you mean by pressure? **[1]**
13. What will be the net force on an object when two forces act on an object in the same direction? **[1]**
14. What is atmospheric pressure? **[1]**
15. Is the gravitational force a contact or non-contact force? **[1]**
16. Define force. **[1]**
17. Name the force due to which every object falls on the earth. **[1]**
18. An inflated balloon was pressed against a wall after it has been rubbed with a piece of synthetic cloth. It was found that the balloon sticks to the wall. What force might be responsible for the attraction between the balloon and the wall? **[1]**
19. What happens when the two forces act in the opposite on an object? **[1]**
20. Do liquids and gases also exert pressure? **[1]**
21. When an objects is thrown vertically upwards why does its speed decrease? **[1]**
22. A ball of dough is rolled into a flat chapatti. Name the force exerted to change the shape of the dough. **[1]**
23. Define the term atmospheric pressure. **[1]**
24. In the given situation identify the agent exerting the force and the object on which it acts. State the effect of the **[1]**

force in case of a load suspended from a spring while its other end is on a hook fixed to a wall.

25. A blacksmith hammers a hot piece of iron while making a tool. How does the force due to hammering affect the piece of iron? **[1]**

26. How do we feel force in our daily life? **[1]**

27. Is force has only magnitude or both direction and magnitude? **[1]**

28. Is the gravity a property of earth only? **[1]**

29. Name the force responsible for the wearing out of bicycle tyres. **[1]**

30. In the given situation identify the agent exerting the force and the object on which it acts. State the effect of the force in case of an athlete making a high jump to clear the bar at a certain height. **[1]**

31. In the given situation identify the agent exerting the force and the object on which it acts. State the effect of the force in case of squeezing a piece of lemon between the fingers to extract its juice. **[1]**

32. What is the site of the pressure exerted by a liquid on the container? **[1]**

33. Do gases also exert pressure on the walls of containers? **[1]**

34. In the given situation identify the agent exerting the force and the object on which it acts. State the effect of the force in case of taking out paste from a toothpaste tube. **[1]**

35. What force will you use to sort out pins easily from garbage? Whether it is a contact force or non-contact force? **[2]**

36. A gas-filled balloon moves up. Is the upward force acting on it larger or smaller than the force of gravity? **[2]**

37. What may be the consequences when a force is applied on an object? **[2]**

38. Explain that liquids exert equal pressure at the same depth. **[2]**

39. During dry weather, clothes made of synthetic fibre often stick to the skin. Which type of force is responsible for this phenomenon? **[2]**

40. A girl is pushing a box towards east direction. In which direction should her friend push the box so that it moves faster in the same direction? **[2]**

41. What is muscular force? Why it is called a contact force? **[2]**

42. Have you ever seen a game of tug-of war? In this game two teams pull at a rope in opposite direction. All members of either team try to pull the rope in their direction. Sometimes the rope simply does not move. Explain why? **[2]**

43. Does force of gravity act on dust particles? **[2]**

44. We know that there is a huge amount of atmospheric pressure on us. But we do not experience its effects why? **[2]**

45. How does an applied force changes the speed of an object? **[2]**

46. Give two examples of situations in which applied force causes a change in the shape of an object. **[2]**

47. Why is force of friction called contact force? **[2]**

48. On what factors does the effect of forces depends? **[2]**

49. A rocket has been fired upwards to launch a satellite in its orbit. Name two forces acting on the rocket immediately after leaving the launching pad. **[2]**

50. While sieving grains, small pieces fall down. Which force pulls them down? **[2]**

51. What is a force? Explain with the help of some examples. **[2]**

52. What happens when **[2]**

    i. Two forces are exerted in same direction?
    ii. Two forces are exerted in opposite directions?

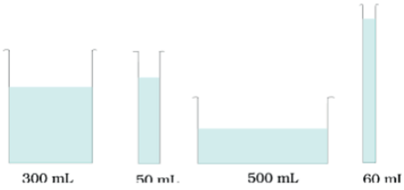
53. Tiny holes are made in a bucket, one near the middle part and the other just above the bottom. When the bucket is filled with water, the water rushes out from the bottom hole much faster than from the upper hole. What **[2]**

conclusion can you observe from this?

54. Give two examples each of situations in which you push or pull to change the state of motion of objects. **[2]**

55. A chapati maker is a machine that converts balls of dough into chapati's. What effect of force comes into play in this process? **[2]**

56. Observe the vessels A, B, C, and D shown in Fig. carefully. **[2]**



Volume of water taken in each vessel is as shown. Arrange them in the order of decreasing pressure at the base of each vessel. Explain.

57. Name the forces acting on a plastic bucket containing water held above ground level in your hand. Discuss why the forces acting on the bucket do not bring a change in its state of motion. **[2]**

58. What do you know by the state of motion? **[2]**

59. Does the force of gravitation exist between two astronauts in space? **[2]**

60. Explain that forces are due to an interaction between objects. **[2]**

61. Define pressure. Write its mathematical expression and SI unit. **[4]**

62. **[4]**
    a. What is meant by atmospheric pressure? What is the cause of atmospheric pressure?
    b. Why are our bodies not crushed by the large pressure exerted by the atmosphere?

63. Explain the working of a dropper and its principle. **[4]**

64. Two women are of the same weight. One wears sandals with pointed heels while the other wears sandals with flat soles. Which one would feel more comfortable while walking on a sandy beach? Give reasons for your answer. **[4]**

65. Explain the atmospheric pressure used in the drinking straw. **[4]**

66. Give reasons: **[4]**
    a. School bags should have wider straps.
    b. A sharp knife cuts better than a blunt knife.

67. What are the various effects of force? **[4]**

68. Explain why, atmospheric pressure decreases as we go higher up above the earth's surface? **[4]**

69. It is easier to walk on soft sand if we have flat shoes rather than the shoes with small heels. Give reason. **[4]**

70. An archer shoots an arrow in the air horizontally. However, after moving some distance, the arrow falls to the ground. Name the initial force that sets the arrow in motion. Explain why the arrow ultimately falls down. **[4]**

71. When a person stands on a cushion, the depression is much more than when he lies down on it. Explain with a reason. **[4]**

72. **[4]**
    a. How does the pressure of a liquid depend on its depth?
    b. Explain why, the walls of a dam are thicker near the bottom than at the top?

73. Name the forces actin on a plastic bucket containing water held above the ground level in your hand. Explain why the forces acting on the bucket do not bring a change in the state of motion. **[4]**

74. **[4]**
    a. What is meant by contact force? Explain with the help of an example.
    b. What is meant by non contact force? Explain with the help of an example.

75. Define electrostatic force. Why it is called a non contact force? Explain by giving examples. **[4]**